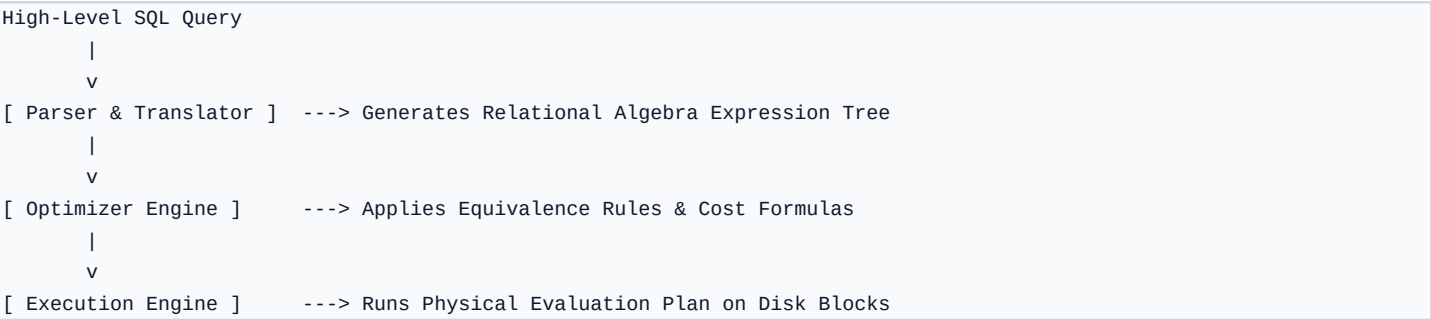


Query Optimization: Transformation of Relational Expressions -- DBMS

Definition: Query Optimization is the component of a Database Management System (DBMS) that attempts to determine the most efficient execution plan for evaluating a given query by transforming relational algebra expressions into equivalent, lower-cost evaluation trees.

1. Detailed Technical Explanation

Query Processing Steps:



Key Relational Algebra Equivalence Rules:

- 1. Commutativity of Selection:

Heuristic Optimization Algorithm:

- 1. Break down complex query conditions into simple selections.

2. Core Concepts & Memory Keywords

- * Equivalence Rules: Algebraic identities ensuring two query trees return identical tuple results.

3. Must-Write Points for Exams

- * Query optimization chooses the physical execution path with minimal disk I/O cost.

4. Quick Recall Flow

SQL Query -> Parse Relational Tree -> Push Selections Down -> Replace Cartesian Product with Join -> Optimal Execution Plan